

R Code for 10 Visualization Scenarios

1. Student Academic Performance

R Code:

```
setwd("Data handling and visualisation")

data <- read.csv("student_performance.csv")

plot(

  data$Attendance,

  data$InternalMarks,

  col = as.factor(data$Department),

  pch = 19,

  xlab = "Attendance (%)",

  ylab = "Internal Marks",

  main = "Student Academic Performance Dashboard"

)

legend(

  "topleft",

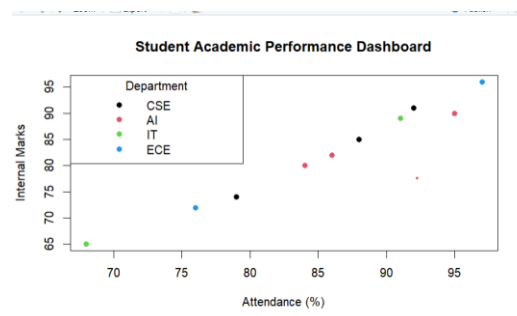
  legend = unique(data$Department),

  col = 1:length(unique(data$Department)),

  pch = 19,

  title = "Department"

)
```



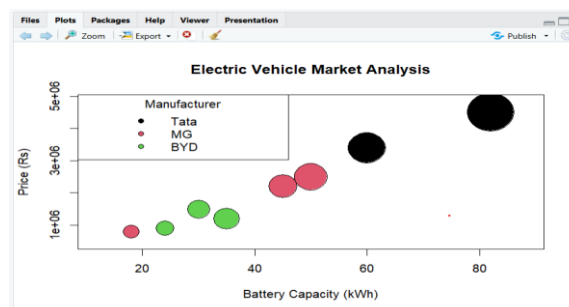
2. EV Market

R Code:

```
data <- read.csv("ev_market.csv")
```

```
symbols(  
  data$BatteryCapacity,  
  data$Price,  
  circles = data$DrivingRange / 50,  
  inches = 0.3,  
  bg = as.numeric(as.factor(data$Manufacturer)),  
  fg = "black",  
  xlab = "Battery Capacity (kWh)",  
  ylab = "Price (Rs)",  
  main = "Electric Vehicle Market Analysis"  
)
```

```
legend(  
  "topleft",  
  legend = unique(data$Manufacturer),  
  pt.bg = 1:length(unique(data$Manufacturer)),  
  pch = 21,  
  title = "Manufacturer"  
)
```



3. Hospital

R Code:

```
data <- read.csv("hospital.csv")

bar <- barplot(

  data$Patients,

  names.arg = data$Department,

  col = rainbow(5),

  main = "Hospital Patient Admission Analysis",

  xlab = "Department",

  ylab = "Number of Patients"

)

text(

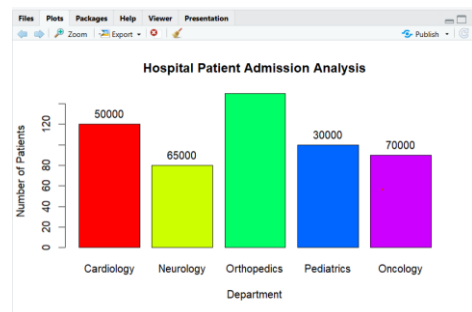
  x = bar,

  y = data$Patients,

  labels = data$TreatmentCost,

  pos = 3

)
```



4. Weather

R Code:

```
data <- read.csv("weather.csv")

chennai <- subset(data, City=="Chennai")

bangalore <- subset(data, City=="Bengaluru")

hyderabad <- subset(data, City=="Hyderabad")

plot(

  chennai$Date,

  chennai$Temperature,

  type="l",
```

```

col="red",

lty=1,

lwd=2,

ylim=c(25,40),

xlab="Date",

ylab="Temperature",

main="Weather Monitoring System"

)

lines(

  bangalore$Date,

  bangalore$Temperature,

  col="blue",

  lty=2,

  lwd=2

)

lines(

  hyderabad$Date,

  hyderabad$Temperature,

  col="green",

  lty=3,

  lwd=2

)

legend(

  "topright",

  legend=c("Chennai","Bengaluru","Hyderabad"),

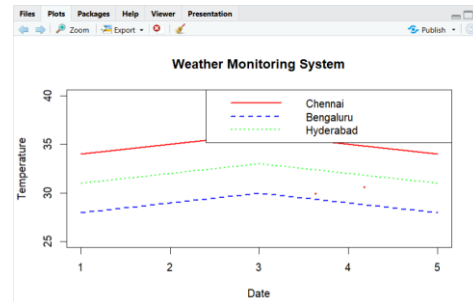
  col=c("red","blue","green"),

  lty=c(1,2,3),

  lwd=2

)

```



5. Sales

R Code:

```
data <- read.csv("sales.csv")

sales <- matrix(

  c(50000,60000,55000,

    35000,40000,37000,

    45000,48000,46000),

  nrow=3,

  byrow=TRUE

)

barplot(

  sales,

  beside=TRUE,

  col=c("red","green","blue"),

  names.arg=c("Food","Drinks","Snacks"),

  main="Supermarket Sales Analysis",

  xlab="Product Category",

  ylab="Sales Amount"

)

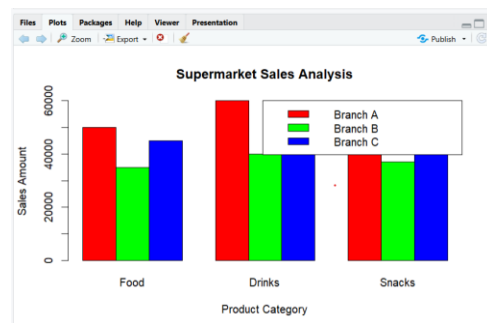
legend(

  "topright",

  legend=c("Branch A","Branch B","Branch C"),

  fill=c("red","green","blue")

)
```



6. Salary

R Code:

```
data <- read.csv("salary.csv")

hist(

  data$Salary,

  breaks=5,
```

```
col="lightblue",

main="Employee Salary Distribution (5 Bins)",

xlab="Salary",

ylab="Frequency"

)
```

```
hist(

data$Salary,

breaks=10,

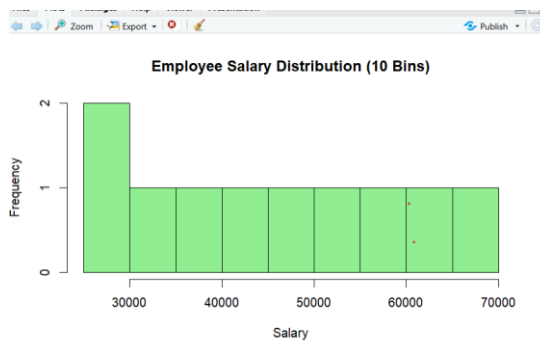
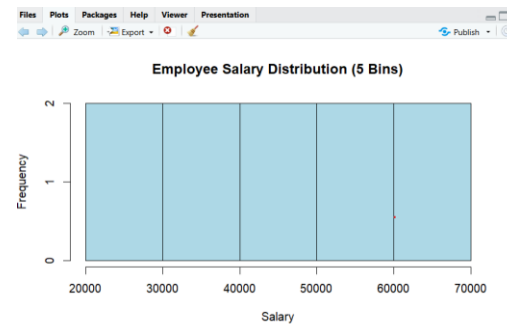
col="lightgreen",
```

```
main="Employee Salary Distribution (10 Bins)",

xlab="Salary",

ylab="Frequency"

)
```



7. Smartphone

R Code:

```
plot(

data$RAM,

data$Price,

pch=as.numeric(as.factor(data$Brand)),

cex=data$Storage/128,

col=heat.colors(nrow(data)),
```

```

main="Smartphone Market Comparison",

xlab="RAM (GB)",

ylab="Price"

)

legend(

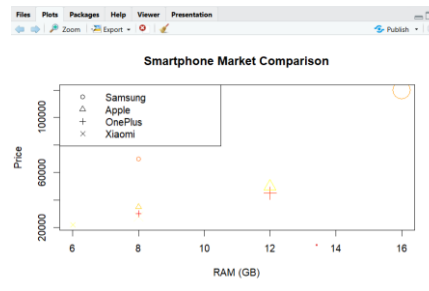
"topleft",

legend=unique(data$Brand),

pch=1:length(unique(data$Brand))

)

```



8. Movie

R Code:

```
data <- read.csv("movie.csv")
```

```
library(ggplot2)
```

```
ggplot(data,
```

```
aes(
```

```
x=Genre,
```

```
y=factor(ReleaseYear),
```

```
fill=AverageRating
```

```
))+
```

```
geom_tile(color="white")+
```

```
scale_fill_gradient(
```

```
low="yellow",
```

```
high="red"
```

```
))+
```

```
labs(
```

```
title="Online Movie Rating Analysis",
```

```
x="Genre",
```

```
y="Release Year",
```

```
fill="Rating"
```

```
)
```



9. COVID

R Code:

```
data <- read.csv("covid.csv")

library(ggplot2)

ggplot(data,

  aes(x = "India",

    y = State,

    fill = ActiveCases)) +

  geom_tile(color = "white") +

  scale_fill_gradient(low = "lightyellow",

    high = "red") +

  labs(

    title = "COVID-19 State-wise Active Cases",

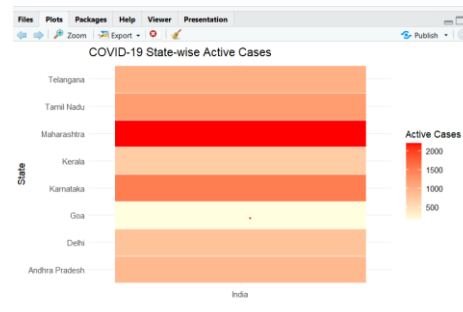
    x = "",

    y = "State",

    fill = "Active Cases"

  ) +

  theme_minimal()
```



10. Iris

R Code:

```
ggplot(

  iris,

  aes(

    x = Sepal.Length,

    y = Petal.Length,

    color = Species,

    shape = Species,

    size = Sepal.Width

  )

) +

  geom_point() +
```

```
labs(
  title = "Iris Flower Classification",
  x = "Sepal Length",
  y = "Petal Length"
) +
theme_minimal()
```

